

Reviewer Pack — Minimal Rank of Twisted Poisson Geometry over SAT Satisfiability

Entry 1f2dbfb39f32 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Twisted Poisson Geometry over SAT Satisfiability

Entry ID: 1f2dbfb39f32 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 22:49:21 UTC

1. Conjecture

Field A (mathematical branch): Twisted Poisson Geometry **Field B** (complexity object): Complexity Theory: SAT Satisfiability

Statement:

{‘text’: ‘For every Boolean satisfiability problem (SAT) instance with n variables and m clauses, the minimal rank of the twisted Poisson manifold associated with its clause indicator matrix is upper-bounded by $2^n * m$.’, ‘invariant’: ‘Minimal Rank of Twisted Poisson Manifold’, ‘object’: ‘Clause Indicator Matrix of SAT Instance’}

Rationale (proposer’s reasoning):

{‘text’: ‘Twisted Poisson geometry has been used to study algebraic structures in dynamical systems, and its minimal rank could potentially capture the complexity of finding satisfying assignments for a Boolean formula. If this conjecture holds, it would suggest a new approach to proving lower bounds on the complexity of SAT.’, ‘linkage’: ‘Linking twisted Poisson geometry to computational complexity’}

Taxonomy category: PoissonGeometry (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 070d85038412527b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of twisted Poisson manifolds associated with clause indicator matrices of SAT instances is upper-bounded by $2^n * m$, where $n \leq 40$ and m clauses are present.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Twisted Poisson Geometry" AND "SAT Satisfiability" - "Minimal Rank" AND "Twisted Poisson Manifold" AND "clause indicator matrix" - "Boolean satisfiability problem" AND $2^n * m$ AND twisted Poisson geometry

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=4.3s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def lcm(a, b):
    return abs(a * b) // gcd(a, b)

def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        if matrix[i][i] == 0:
            for j in range(i + 1, n):
                if matrix[j][i] != 0:
                    matrix[i], matrix[j] = matrix[j], matrix[i]
                    break
            else:
                raise ValueError("Matrix is singular")
        factor = Fraction(matrix[i][i])
        for j in range(n):
            matrix[i][j] /= factor
        for j in range(n):
            if i != j:
                factor = Fraction(matrix[j][i])
                for k in range(n):
                    matrix[j][k] -= factor * matrix[i][k]
    return matrix

def construct_twisted_poisson_matrix(variables, clauses):
```

```

n = len(variables)
m = len(clauses)
matrix = [[0] * (n + m) for _ in range(n + m)]

for i in range(n):
    matrix[i][i] = 1

for j in range(m):
    var1, var2 = clauses[j]
    if var1 < n:
        matrix[var1][n + j] = -1
    else:
        matrix[n + var1 - n][n + j] = -1

    if var2 < n:
        matrix[var2][n + j] = 1
    else:
        matrix[n + var2 - n][n + j] = 1

return gaussian_elimination(matrix)

def run_trial(seed: int) -> dict:
    random.seed(seed)

    n = random.randint(5, 40)
    m = random.randint(n, n * 3)
    variables = list(range(n))
    clauses = [(random.choice(variables), random.choice(variables)) for _ in range(m)]

    try:
        matrix = construct_twisted_poisson_matrix(variables, clauses)
        rank = sum(1 for row in matrix if any(row[i] != 0 for i in range(len(row))))

        expected_bound = 2 ** n * m
        conjecture_holds = rank <= expected_bound

        return {
            "metric_name": "Minimal Rank of Twisted Poisson Manifold",
            "metric_value": rank,
            "instances_tested": 1,
            "conjecture_holds": conjecture_holds,
            "counterexample": "" if conjecture_holds else f"Rank {rank} > {expected_bound}"
        }
    except Exception as e:
        return {
            "metric_name": "Minimal Rank of Twisted Poisson Manifold",
            "metric_value": None,
            "instances_tested": 1,
            "conjecture_holds": False,
            "counterexample": str(e)
        }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else list(range(2, 30)) + list(range(100, 1000))

```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

total_rank = sum(r["metric_value"] for r in results if r["metric_value"] is not None)
num_tests = sum(r["instances_tested"] for r in results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(result["conjecture_holds"] for result in results):
    print(f"RESULT: SUPPORTED mean={total_rank / num_tests:.2f} std=0.00 support_fraction=1.00")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={total_rank / num_tests:.2f} std=0.00 support_fraction={support_fraction:.2f}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample='{result['counterexample']}' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ed Poisson Manifold', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': None
TRIAL: {'metric_name': 'Minimal Rank of Twisted Poisson Manifold', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank of Twisted Poisson Manifold', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': None}
TRIAL: {'metric_name': 'Minimal Rank of Twisted Poisson Manifold', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': None}
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TRIAL: {'metric_name': 'Minimal Rank of Twisted Poisson Manifold', 'metric_value': None, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': None}
RESULT: FALSIFIED counterexample='Matrix is singular' first_failing_seed=11

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test has only been conducted on a single instance, which is insufficient to draw a general conclusion. The counterexample provided ('Matrix is singular') does not necessarily refute the conjecture; it could be an artifact of the specific instance chosen or the method used to compute the minimal rank.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test has provided a counterexample where the matrix associated with a SAT instance is singular, indicating that the minimal rank of the twisted Po | next: Further investigation is needed to determine if this counterexample is representative of all SAT instances or if it is an artifact of the specific

instance chosen or the method used. Additional testing with a variety of SAT instances and different methods for computing the minimal rank sh

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12507
2	preregistration	ollama_remote	glm4:latest	0	0	5702
3	novelty	ollama_remote	glm4:latest	0	0	4714
4	novelty	ollama_remote	glm4:latest	0	0	9534
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14482
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11707
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15971
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11473
9	critic	ollama_remote	glm4:latest	0	0	12031
10	judge	ollama_remote	glm4:latest	0	0	5919

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 104040 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/1f2dbfb39f32.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/1f2dbfb39f32.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/1f2dbfb39f32.tar.gz* (if generated)